

Ziqing Li

📞 (+86) 13955537233 @ d202381502@hust.edu.cn

🏢 Storage laboratory at Institute of Wuhan National Research Center for Optoelectronics

🏢 School of Computer Science and Technology 🏛️ Huazhong University of Science and Technology

🎓 Education

2023 - present	PhD student in Computer Storage System, HUST
2019 - 2023	B.S. in Computer Science and Technology, Beijing Jiaotong University

⚙️ Research experience

LLM Inference Acceleration with PCIe 5.0 SSD Arrays Apr. 2025 - Present

Advisor: **Assoc. Prof. Jianxi Chen & Prof. Dan Feng**, School of Computer Science and Technology, HUST

- Designed a hotness-aware tiering algorithm for online KV block classification, P99 latency < 60μs
- Built a GPU-DRAM-NVM-SSD four-tier hierarchy atop OrchFS, routing warm data via NVM 4KB pages and cold data via 32KB SSD blocks, boosting SSD write bandwidth from 4% to 41%+
- Implemented a compute-prefetch-preload pipeline to hide migration latency, achieving bit-exact output vs. GPU-only baseline with < 0.5% throughput overhead

GPU-Direct Storage for GPU-SSD Collaborative Systems Mar. 2026 - Present

Advisor: **Assoc. Prof. Jianxi Chen & Prof. Dan Feng**, School of Computer Science and Technology, HUST

- Explore GPU Direct Storage (GDS) for direct GPU-SSD data transfer, bypassing CPU and host memory
- Analyze GPU-SSD data-movement bottlenecks and storage/runtime co-design for LLM serving

Stock Prediction with Traditional and Alternative Financial Data Jul. 2022 - Oct. 2022

Advisor: **Prof. Guangnan Ye**, Fudan Vision and Learning Laboratory, FDU

- Built stock prediction models with TOPNet, LSTM, and LightGBM, and explored financial text and user behavior data using NLP and graph-based models

💻 Work experience

Big data computing engine development Nov. 2023 - Feb. 2024

Advisor: **Senior SDE, Apache committer Yaodong Zhang**, Data Center Department, OLAP Engine Group, Xiaomi

- Implemented Parquet-level encryption and decryption modules within Spark, enabling column-level data protection for internal OLAP pipelines

Federated learning in medical image processing Aug. 2022 - Sep. 2022

Advisor: **Asst. Prof. Chenghao Cai**, Artificial Intelligence Center, NUS Research Institute (Su Zhou)

- Benchmarked FedAvg and FedProx for medical image classification under non-IID distributions

🏆 Awards

- **Huazhong University of Science and Technology First-class academic scholarship** Sep. 2023-25
- **Merit Student Award of Beijing Jiaotong University** Dec. 2020
- **Beijing Jiaotong University Excellent Second Class Scholarship** Dec. 2021

🔧 Skills

📄 Programming	C/C++, Java, Python, Scala, Matlab
📁 Systems & Tools	Git, GDB, SPDK, GDS, HDFS
📖 Research Areas	Storage Systems, LLM Inference Optimization, RDMA